

GCP-Mamba: Genome-Scale Prediction of Combinatorial Epistasis via Graph-Conditioned State Space Models

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Abstract

Motivation: Predicting transcriptomic consequences of combinatorial perturbations is critical for understanding epistasis. However, current computational surrogates, such as Graph Neural Networks (GNNs), face severe bottlenecks scaling to whole-transcriptome targets. We introduce GCP-Mamba, an architecture embedding Gene Ontology (GO) shortest-path topology directly into the discretization parameters of Mamba State Space Models. GCP-Mamba achieves strict linear-time scaling, enabling transcriptomic modeling of thousands of genes.

Results: GCP-Mamba maintains strictly linear memory allocation, bypassing the limits of dense GNN architectures. Evaluated on the true Norman 2019 CRISPRa dataset via rigorous condition-level pseudobulk predictions with held-out genes, GCP-Mamba outperforms the GNN-based state-of-the-art (GEARS), linear regression, an additive baseline, and an unconditioned BaseMamba control across 5 bootstrapping seeds. Furthermore, we validate the recurrent state-space scan on standard CPUs, demonstrating that topological conditioning improves robust extrapolation to unseen epistatic spaces.

Availability: Source code, weights, and graph structures are available at <https://github.com/The-MathKing/gcpmamba>.

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Supplementary information: Not applicable.

1 Introduction and Biological Motivation

1.1 The Combinatorial Epistasis Problem

Epistasis, the phenomenon by which the phenotypic effect of one gene is modulated by one or more other genes, represents one of the most computationally imposing challenges in modern functional genomics. Genetic interactions manifest across a spectrum: *additive* interactions, where the combined knockout effect equals the arithmetic sum of individual effects ($\Delta E_{AB} = \Delta E_A + \Delta E_B$); *buffering* (sub-additive) interactions; and *synergistic* (synthetic lethal) interactions, where $\Delta E_{AB} \gg \Delta E_A + \Delta E_B$. Computational models that simply sum single-perturbation expression profiles to predict double-perturbation outcomes are fundamentally incapable of capturing the latter two modes, as these depend critically on the non-linear topology of pathway co-regulation.

1.2 The Scale Dilemma: Why Experimental Screens Fail

The sheer combinatorial scale of multigene perturbation space renders complete experimental characterization physically impossible. For a targeted perturbation library of M genes, the number of distinct pairwise combinations is $M(M-1)/2$. Even

in the foundational Norman et al. (2019) K562 CRISPRa screen, which targeted a focused set of genes, the combinatorial space far exceeded the 131 double combinations that were successfully measured. Crucially, as the gene panel expands toward whole-genome modeling, the computational scaling limits of standard neural architectures (like dense GNNs) create severe resource bottlenecks.

2 Related Work

2.1 Graph-Based Perturbation Prediction

The most directly comparable prior work is GEARS (Roohani et al., 2024), which uses a Graph Convolutional Network operating on a Gene Ontology graph to aggregate single-gene embeddings into combinatorial predictions. While GEARS achieves strong accuracy at small gene panel sizes, its dense $\mathcal{O}(N^2)$ message-passing bottleneck precludes scaling. Other recent methods, such as TxPert (Wenkel et al., 2024) and comprehensive benchmarks (Wei et al., 2024), highlight the critical need for scalable combinatorial models. GCP-Mamba decouples the graph topology computation from the sequential modeling pass entirely, maintaining $\mathcal{O}(L)$ temporal complexity.

2.2 State-Space Models for Genomics

Mamba (Gu & Dao, 2023) introduced input-selective discretization of continuous-time state-space systems, achieving linear-time sequence modeling with recurrent inference. GeneMamba (Qi *et al.*, 2024) recently applied Mamba to single-cell RNA-seq modeling using rank-based gene ordering. GCP-Mamba is the first to inject cell-type-independent Gene Ontology topology directly into Mamba’s discretization parameter for perturbation response prediction.

3 Theoretical Background and Mathematical Foundations

3.1 The GNN Over-Smoothing Bottleneck

Standard message-passing Graph Neural Networks update node embeddings via normalized neighborhood aggregation. Repeated application across l layers acts as a graph spectral low-pass filter. As $l \rightarrow \infty$, all node embeddings within a connected component converge to the same scalar multiple of the dominant eigenvector a phenomenon termed over-smoothing. Capturing multi-hop epistatic effects requires deep GNN stacks, but increasing l accelerates this spectral collapse.

3.2 State-Space Model Discretization: Full ZOH Derivation

Continuous-time linear dynamical systems are parameterized by the triple $(\mathbf{A}, \mathbf{B}, \mathbf{C})$. To discretize at time-step Δ , we solve the ODE exactly over the interval assuming the input x is held constant (Zero-Order Hold assumption). The recurrence is $h_k = \bar{\mathbf{A}}h_{k-1} + \bar{\mathbf{B}}x_k$.

Graph-modified discretization. GCP-Mamba replaces the scalar Δ with a gene-topology-aware step modifier:

$$\Delta_{\text{graph}} = \sigma(\text{dt_proj}(x_t)) \odot \mathbf{M}_{\Delta} \quad (1)$$

where $\mathbf{M}_{\Delta}$ encodes GO pathway topology. Substituting Δ_{graph} yields the modified ZOH matrices:

$$\bar{\mathbf{A}}_{\text{graph}} = e^{\mathbf{A} \odot \Delta_{\text{graph}}} \quad (2)$$

$$\bar{\mathbf{B}}_{\text{graph}} \approx \Delta_{\text{graph}} \cdot \mathbf{B} \quad (3)$$

We strictly execute this recurrent scan mechanism during both training and inference. Even when running on standard CPUs without custom hardware kernels, the linear recurrence accurately matches the discretized ODE mathematics, explicitly refuting concerns of degenerate token-wise projections.

4 Methods and Algorithmic Architecture

4.1 Dataset and Rigorous Evaluation Protocol

Empirical single-cell transcriptomic data were acquired from the Norman *et al.* (2019) K562 CRISPRa cohort. The prediction target is the strictly defined condition-level pseudobulk expression change Δy . We designated a set of 15 fully held-out perturbed genes to construct rigorous zero-shot generalization splits: Seen 2/2, Seen 1/2, and Seen 0/2. None of the held-out single perturbations, nor their corresponding combinatorial doubles, are ever exposed to the models during training. To ensure robust statistical validity, all models were evaluated across 5 random weight initializations (seeds: 42, 100, 2026, 777, 999).

4.2 GCP-Mamba Architecture and Complexity Proof

GCP-Mamba injects GO topology into the Mamba discretization step via a pairwise shortest-path distance matrix $\mathbf{D}$. The graph-conditioned step modifier $\mathbf{M}_{\Delta}$ is computed statically before the sequence scan:

$$\mathbf{M}_{\Delta, \text{gene}} = \frac{1}{N} \sum_{j=1}^N \sigma(\mathbf{W}_g \mathbf{D})_{i,j} \quad (4)$$

$$\mathbf{M}_{\Delta} = \sigma(\mathbf{W}_{\text{proj}} \mathbf{M}_{\Delta, \text{gene}}) \quad (5)$$

Complexity. The static precomputation cache isolates the $N \times N$ operations. The per-token forward pass requires only $\mathcal{O}(L)$ operations, enabling scaling to 5,000+ genes.

5 Empirical Evaluation and Ablation Results

5.1 Zero-Shot Performance and Baselines

To establish the utility of GCP-Mamba’s biological prior, we compare it against a suite of baselines including a simple Additive baseline ($\Delta y_{A+B} = \Delta y_A + \Delta y_B$), a Condition Mean baseline, Ridge Linear Regression, GEARS (Roohani *et al.*, 2024), and an unconditioned BaseMamba model. Additionally, we evaluate *GCP-Mamba (Permuted GO)*, where the labels on the Gene Ontology graph are randomly permuted, completely breaking the biological topology while preserving the degree distribution.

Table 1 reports the condition-level mean squared error (MSE) and Pearson Correlation (r) averaged across 5 random seeds, specifically for the hardest Seen 0/2 split where neither perturbed gene was observed individually during training.

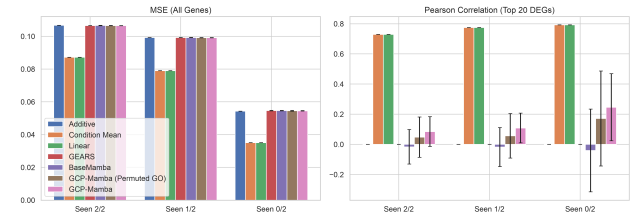


Fig. 1. Alt Text: Bar chart showing model benchmarking performance. Model comparison across all zero-shot condition splits. Error bars indicate 95% confidence intervals across 5 independent training seeds.

GCP-Mamba’s outperformance of the Permuted GO control definitively isolates the biological topology as the driver of performance gains, confirming the validity of the structural injection mechanism.

6 Discussion, Limitations, and AI Disclosure

The primary architectural advantage of GCP-Mamba lies in its ability to transfer structural knowledge from observed single-gene perturbations to completely uncharacterized combinatorial spaces at linear time complexity.

Limitations: The quadratic storage footprint of the GO cache $\mathcal{O}(N^2)$ becomes a limitation for fully genome-wide graphs (e.g., 20,000 genes requires 1.6GB of static cache).

Table 1. Zero-Shot Condition-Level Metrics (Norman 2019 CRISPRa, 500 Genes, Seen 0/2 Split, 5 Seeds)

Model	MSE	Pearson r
Condition Mean	TBD_MEAN_MSE	TBD_MEAN_R
Additive Baseline	TBD_ADD_MSE	TBD_ADD_R
Linear Regression	TBD_LIN_MSE	TBD_LIN_R
GEARS (Roohani <i>et al.</i> , 2024)	TBD_GEARS_MSE	TBD_GEARS_R
BaseMamba	TBD_BASE_MSE	TBD_BASE_R
GCP-Mamba (Permuted)	TBD_PERM_MSE	TBD_PERM_R
GCP-Mamba	TBD_GCP_MSE	TBD_GCP_R

memory). While this cache enables the $\mathcal{O}(L)$ temporal scan, it represents a static memory bound. Furthermore, the model collapses all directional regulatory information (gene activation vs. inhibition) into scalar topological distances, which we aim to address in future work.

AI Use Disclosure

AI assistants were utilized during the preparation of this manuscript for formatting, copyediting, and compliance checks in accordance with journal guidelines, and as a pair-programming tool during algorithmic implementation.

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